

# Linux Security



# User & Permission Model

- Linux uses users, groups, and permissions to control access.
- Each file has Read (r), Write (w), Execute (x) permissions.
- Root user has full system control.
- Principle of Least Privilege improves system security.

# **SSH** *(Secure Shell)*

- SSH allows secure remote login into Linux systems.
- Uses encryption to protect data.
- Password-based login is risky.
- Key-based authentication is more secure.

# **Firewall**

## ***(IPTables / UFW)***

- Firewall controls incoming and outgoing network traffic.
- IPTables works using rules.
- Blocks unauthorized access to services.
- UFW is a user-friendly firewall interface.

 **Fail2Ban**



- Protects system from brute-force attacks.
- Monitors log files (SSH, FTP, etc).
- Automatically blocks attacker IPs.
- Uses temporary or permanent bans.



# System Logs & Monitoring

- Logs help detect suspicious activity.
- `/var/log/auth.log` stores authentication events.
- Continuous monitoring helps in early attack detection.

# Linux Hardening

- Disable unused services.
- Use strong passwords.
- Keep system updated.
- Limit root access.
- Enable firewall and monitoring.